

FullStack Development – Summary

Today covered topics:

- ❖ Array
- ❖ Types of array
- ❖ Jagged array

➤ Array:

- Array is a collection of element in same datatype stored in memory.
- Easy to manage , loop and process. Index starts with 0.
- Array size is fixed.

Syntax:

```
datatype[] arrayname = new datatype[size];
```

Advantage:

- ✓ Iteration – repetitive loop.
- ✓ Code simplicity.
- ✓ Memory efficiency
- ✓ Easy Access.
- ✓ Sorting/searches.

➤ Types of array:

1. One dimensional array:

One dimensional array is a single row of elements.

Eg:

```
public class Single {  
  
    public static void main(String[] args) {  
        int[] num = {5, 10, 15, 20};  
        int sum = 0;  
  
        for (int i = 0; i < num.length; i++) {  
            sum += num[i];  
        }  
  
        System.out.println("sum: " + sum);  
    }  
}
```

2.Two dimensional Array:

Its a array of array and then its look like matrix structure and table format it can store data in column form.it required nested loops.

Syntax:

```
datatype[][] array=new datatype [rows][columns];
```

Eg:

```
public class Twodimensional {  
  
    public static void main(String[] args) {  
  
        int[][] Array = {  
            {1, 2, 3},  
            {4, 5, 5},  
            {3, 4, 5}  
        };  
  
        for (int i = 0; i < 3; i++) {  
            for (int j = 0; j < 3; j++) {  
                System.out.print(Array[i][j] + " ");  
            }  
            System.out.println();  
        }  
    }  
}
```

➤ **Jagged Array:**

- Is a 2-D array each row can have a different number of columns.
- In 2-D array have a row in same length.
- Jagged array have a row different length.

Syntax:

```
Datatype[][] arrayname=new datatype[number of rows][];
```

Advantage:

- 1.flexible structure
- 2.better efficiency
- 3.save memory
- 4.used for real time

Eg:

```
public class JaggedArrayExample {  
  
    public static void main(String[] args) {  
  
        int[][] arr = {  
            {10, 20},
```

```
        {30, 40, 50},  
        {60, 70, 80, 90}  
    };  
  
    for (int i = 0; i < arr.length; i++) {  
        for (int j = 0; j < arr[i].length; j++) {  
            System.out.print(arr[i][j] + " ");  
        }  
        System.out.println();  
    }  
}
```